

MOTIVATION	DATA COLLECTION	PREPROCESSING	EXPLORATORY DATA ANALYSIS (EDA)	VISUALIZATIONS
Most people cook at home and herbs and spices are an important part in the foods flavor. Unless you are following a recipe or already have some knowledge of what goes well together, trying out new herb/spice pairings might be difficult or you end up with a combination that does not pair well together. What spices you should use also depend on what vegetables, meats etc. you are using, and what flavor profile fits your taste. My idea is to make a simple web page we're the user can easily see what herbs go well together and be able to limit the options to spices that they already own. This would hopefully help with finding new flavor blends using just the spices you already have or finding new things to add to your spice cabinet/drawer. The user could also select a base food ingredient and get spice recommendations for it, or select a cuisine (french, japanese, etc) and get recommendations that with that flavor profile.	I am using a Food herb & spice pairing chart by Pennsylvania State University as my base data and add information from other public sources to it. Most of this data will be just website scraping. If I have time or if needed, maybe adding recipe data could be interesting too.	I am compiling all the data into a database format myself. I still need to figure out what the best way is to format all the information, so that it is easy and logical to get the comparison between different spices and what foods they pair well with. If/When I am to add data from multiple sources, I am probably going to have to deal with some missing data. The data I am mainly using also doesn't show all the two way relationships between herbs. I am going to be working with purely categorical data that I need to transform into numerical form, like using one-hot encoding.	Since I am already working with limited data that I have to "gather by hand", my first instinct is that the the EDA process will already be close to my final visualisation process. The main concern is to figure out what is the best way(s) to easily visualise the relationships between the different combinations. Ofcourse I will most likely have other ideas once I have combined the data.	The idea is to have an interactive visualisation. Since the project will most likely be lacking in ML methods etc. I feel like this is the most crucial part of this project. The user should be able to see the maybe a circular node graph and be able to interact with it. The user should also be able to choose specific herbs and get "recommended" which of those herbs pair well together and what foods they would work well with. I don't know if it is needed to also have some base statistics about the data, like what herb mixes best with the most ingredients or what foods are underrepresented in the data.

LEARNING TASK (focus on problem definition) The learning part would come from the user being able to input a base food product (vegetable, meat) and add a flavor/taste profile) and get spice recommendations for the food or vice versa the user could pick herbs they have and then get recommended what dishes you could use some of those spices on. I will try seeing if the flavor/taste of the herbs can be used to predict what spices work well together and find some way to rank them (maybe more herbs that fit → higher recommendation for starters).	LEARNING APPROACH (focus on solution implementation) I am unsure still which ML methods or evaluations I should use. Maybe clustering could help visualise the data		COMMUNICATION OF RESULTS All results should be conveyed in the interactive web page. A box for graphs and some fields or checklist for spice, flavor, food and cuisine selection	DATA PRIVACY AND ETHICAL CONSIDERATIONS <i>(if applicable)</i> All data used should be generally public information.
	ADDED VALUE I want to see if we can also predict some combinations not seen in the data explicitly.			LEGEND WEEK 1: Data collection/preprocessing WEEK 2: EDA & visualizations WEEKS 3-4: Machine/deep learning WEEK 5: Fairness & data privacy